

1. Suppose f is such that $f(-x) = -f(x)$ for every real x and $\int_0^1 f(x) dx = 5$, then $\int_{-1}^0 f(t) dt =$
 (a) 10 (b) 5 (c) 0 (d) -5
2. Area bounded by the curves $4y = |x^2 - 4|$ and $y + |x| = 7$, is equal to
 (a) 8 sq. units (b) 16 sq. units
 (c) 4 sq. units (d) 32 sq. unit
3. $\int_{\pi/4}^{3\pi/4} \frac{dx}{1 + \cos x}$ is equal to
 (a) 2 (b) -2 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$
4. The value of $\int_{-1}^1 \frac{d}{dx} \left(\tan^{-1} \frac{1}{x} \right) dx$ is
 (a) $\pi/2$ (b) $\pi/4$ (c) $-\pi/2$ (d) None of these
5. The area enclosed by the curves $3x^2 + 5y = 32$ and $y = |x - 2|$ is -
 (a) $\frac{17}{2}$ (b) $\frac{33}{2}$ (c) $\frac{23}{2}$ (d) None of these
6. A point $P(x, y)$ moves in xy plane in such a way that $\sqrt{2} \leq |x + y| + |x - y| \leq 2\sqrt{2}$. Area of the region representing all possible positions of the point 'P', is equal to
 (a) 2 sq. units (b) 4 sq. units
 (c) 6 sq. units (d) 8 sq. unit
7. Let $f(x) = x - [x]$, for every real number x , where $[x]$ is integral part of x . Then $\int_{-1}^1 f(x) dx$ is.
 (a) 1 (b) 2 (c) 0 (d) $1/2$
8. Let f, g and h be continuous function on $[0, a]$ such that $f(x) = f(a - x)$, $g(x) = -g(a - x)$ and $3h(x) - 4h(a - x) = 5$.
 Then $\int_0^a f(x) g(x) h(x) dx =$
 (a) $5/4$ (b) $3/4$ (c) 1 (d) 0
9. $\int_0^{\log_5 e^x \sqrt{e^x - 1}} \frac{e^x \sqrt{e^x - 1}}{e^x + 3} dx =$
 (a) $3 + 2\pi$ (b) $4 - \pi$ (c) $2 + \pi$ (d) $4 + \pi$
10. $\int_{-2}^2 \sin^3 x |x| dx$ is equal to -
 (a) 0 (b) 1 (c) 4 (d) None of these
11. $\int_{-1}^1 \frac{1}{1 + x^2} dx =$
 (a) π (b) $\frac{\pi}{2}$ (c) $-\frac{\pi}{2}$ (d) $\frac{\pi}{3}$
12. $\int_0^{\pi/4} (\sqrt{\tan x} + \sqrt{\cot x}) dx$ is equal to
 (a) $\frac{\sqrt{2}\pi}{2}$ (b) $\frac{\pi}{2}$ (c) $\frac{\sqrt{3}\pi}{2}$ (d) None of these
13. $\int_{-\pi/2}^{\pi/2} \sin^{10} x (6x^9 - 25x^7 + 4x^3 - 2x) dx$ equals to :

- (a) π (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) None of these
14. Find the area bounded by $x = 1/2$, $x = 2$, $y = \log_e x$ and $y = 2^x$ -
 (a) $\frac{4-\sqrt{2}}{\log 2} + \frac{5}{2} \log 2 + \frac{3}{2}$ (b) $\frac{4-\sqrt{2}}{\log 2} - \frac{5}{2} \log 2 + \frac{3}{2}$
 (c) $\frac{4+\sqrt{2}}{\log 2} - \frac{5}{2} \log 2 + \frac{3}{2}$ (d) None
15. The area of the bounded by $y = \cos x$, $y = 0$, $|x| = 1$ is given by -
 (a) $\sin 1$ (b) $2 \sin 1$ (c) $4 \sin 1$ (d) None of these
16. The area of the region bounded by $y = e^x$, $y = e^{-x}$, $x = 0$ and $x = 1$ is -
 (a) $e + \frac{1}{e}$ (b) $\log(4/e)$ (c) $4 \log(4/e)$ (d) $e + \frac{1}{e} - 2$
17. Area lying between the curves $y = \tan x$, $y = \cot x$ and x -axis, $x \in [0, \pi/2]$ is
 (a) $\log 2$ (b) $\frac{1}{2} \log 2$ (c) $2 \log \left(\frac{1}{\sqrt{2}} \right)$ (d) None of these
18. $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ equals
 (a) $\pi/2$ (b) $\pi/4$ (c) π (d) 2π
19. $\lim_{x \rightarrow 0} \frac{1}{x} \left(\int_y^a e^{\sin^2 t} dt - \int_{x+y}^a e^{\sin^2 t} dt \right) =$
 (a) 1 (b) 0 (c) $e^{\sin^2 y}$ (d) $\sin 2y \cdot e^{\sin^2 y}$
20. The area of the region bounded by the curve
 $y = \frac{16-x^2}{4}$ and $y = \sec^{-1}[-\sin^2 x]$ (where $[.]$ denotes the greatest integer function) is -
 (a) $\frac{1}{3} (4 - \pi)^{3/2}$ (b) $8(4 - \pi)^{3/2}$
 (c) $\frac{8}{3} (4 - \pi)^{3/2}$ (d) $\frac{8}{3} (4 - \pi)^{1/2}$